

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0693

Fumarole 138 kV
May 01, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0693. The Interconnection Customer proposes a 121.5 MW solar photovoltaic facility interconnecting at the Fumarole 138 kV substation in the Virginia area. The request was received on October 24, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 277 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,591,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ARTSIA COUNTRY CLUB 345/1kV Transformer	ARTESIA COUNTRY CLUB / AIR FORCE BASE 345kV	108.9	540	8.3
ROSS FORK / ARTESIA COUNTRY CLUB 138kV line	AIR FORCE BASE / ARTESIA COUNTRY CLUB 345kV	102.9	647	8.1
ARTESIA COUNTRY CLUB 138/1 kV XFMR	ARTESIA COUNTRY CLUB / AIR FORCE BASE 345kV L	102.3	575	8.3
MONOLITH / SANTA ROSA ENERGY CTR 138kV L	INTERNATIONAL PAPER VALLIANT OK / KEARNEYVILLE 345kV Line	101.6	252	8.0
ABBOTSFORD / LE GRAND 138kV line	W WINCHESTER / INTERNATIONAL PAPER VALLIANT OK 138kV	101.4	267	8.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ARTSIA COUNTRY CLUB 345/1kV Transformer	Replace transformer with higher-rated unit	\$13,909,000	9% (\$1,275,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ROSS FORK / ARTESIA COUNTRY CLUB 138kV line	Reconductor line	\$1,396,000	7% (\$100,000)
ARTESIA COUNTRY CLUB 138/1 kV XFMR	Replace transformer with higher-rated unit	\$6,901,000	9% (\$632,000)
SANTA ROSA ENERGY CTR / MONOLITH 138kV	Reconductor line	\$1,242,000	28% (\$351,000)
ABBOTSFORD / LE GRAND 138kV line	Reconductor line	\$3,191,000	70% (\$2,233,000)
	Total allocated to project		\$4,591,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.